A STORY IN NORTHWEST COAST

Feasible and Cost-effective Solutions to Safeguard Singapore's Northwest Coast

by Team Roller Coastal



INTRODUCTION

Team Roller Coastal

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE



Chai Jin Xie

Year 4
Civil
Engineering



Yang Ze Wen

Year 4
Civil
Engineering



Si Poh Yi

Year 3
Civil
Engineering



Teh Jun Tao

Year 2
Civil
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Yap Kai Xiang

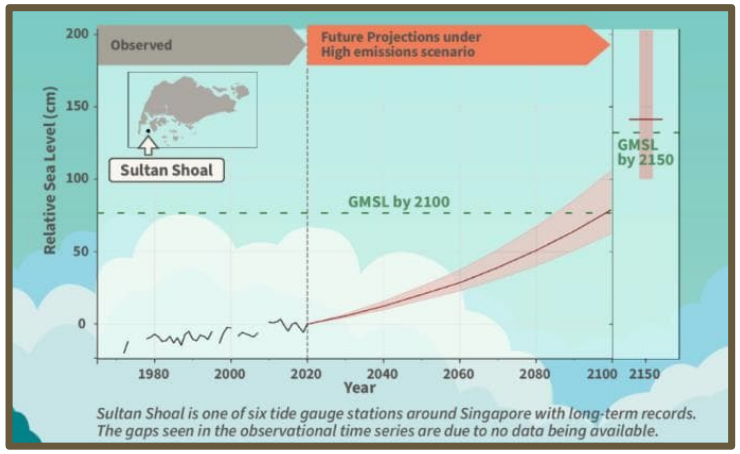
Year 2
Civil
Engineering



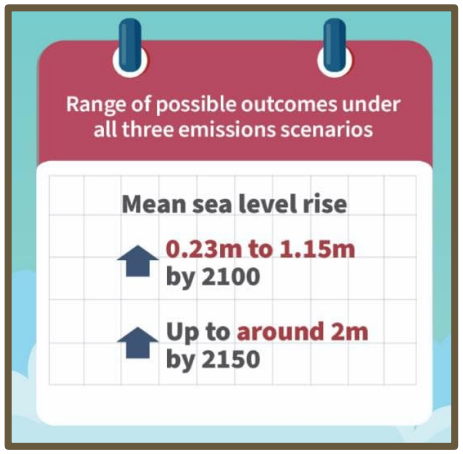
BACKGROUND

01

RISING SEA LEVEL



Source: *https://www.nccs.gov.sg/images/v3_infographic_english.jpg?sfvrsn=1e966f02_0*



CNN Weather Video Climate

Flooding from sea level rise could cost our planet $14.2 trillion, study says

By Helen Regan, CNN

5 minute read · Published 11:02 AM EDT, Thu July 30, 2020

(CNN) — If the world does nothing to mitigate sea level rise, coastal flooding will become so extreme and destructive that it could cause damage worth up to 20% of global gross domestic product by 2100, according to new research.

The study, published Thursday in the journal Science Reports, is the first to map the projected economic impact of sea level rise caused by the climate crisis.

The authors say rising sea levels could cost the global economy $14.2 trillion in lost or damaged assets by the end of the century, as larger areas of land, home to millions of people, are inundated.

Source: *CNN, https://edition.cnn.com/2020/07/30/weather/coastal-flooding-sea-level-rise-study-intl-hnk/index.html*

FUNDING



THE STRAITS TIMES

National Day Rally 2019: $100 billion needed to protect Singapore against rising sea levels

SINGAPORE - Around $100 billion or more may be needed over the long term to protect Singapore against rising sea levels, one of the many threats posed by climate change and one which the country is especially vulnerable to, said Prime Minister Lee Hsien Loong on Sunday (Aug 18).

There are good engineering solutions to the problem and these could include reclaiming offshore islands and connecting them with barrages, or building polders to protect the coastline. But they come at a cost.

"How much will it cost, to protect ourselves against rising sea levels? My guess is probably $100 billion over 100 years, quite possibly more.

Source: The Straits Time, https://www.straitstimes.com/singapore/national-day-rally-2019-100-billion-needed-to-protect-singapore-against-rising-sea-levels



THE STRAITS TIMES

Singapore Budget 2020: New coastal and flood protection fund to protect Singapore against rising sea levels

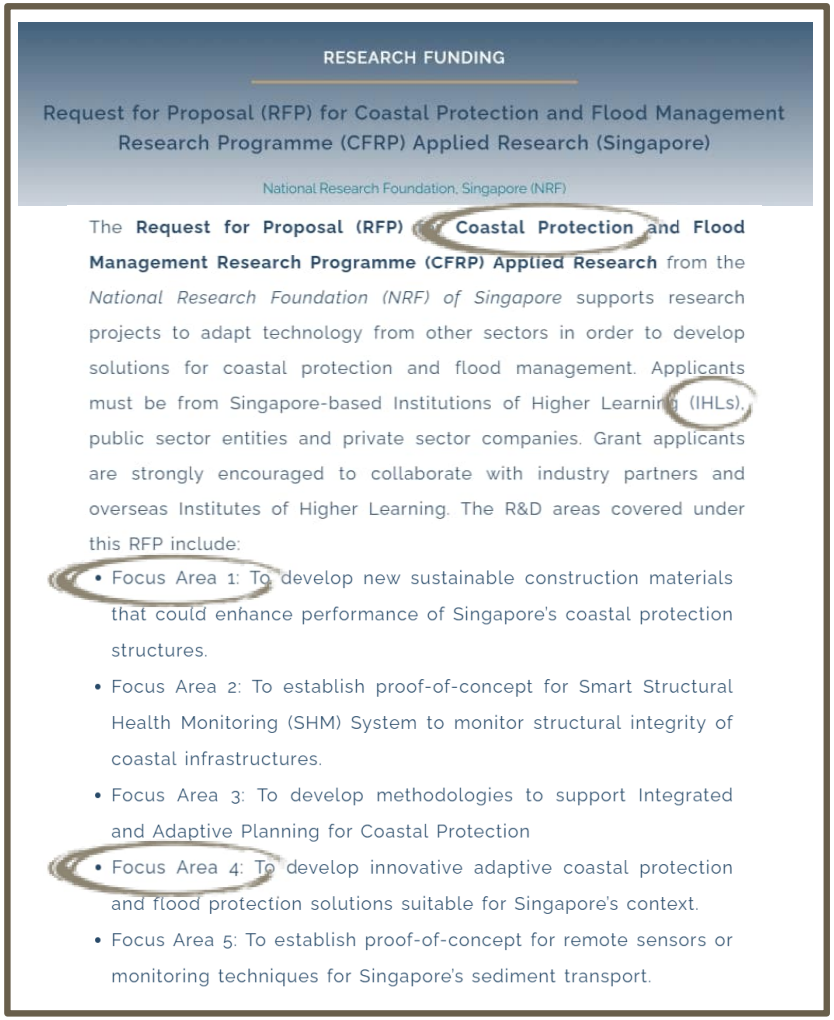
SINGAPORE - A new coastal and flood protection fund, with an initial injection of $5 billion, will be set up to help protect Singapore against rising sea levels, Deputy Prime Minister Heng Swee Keat said on Tuesday (Feb 18).

The fund will be topped up subsequently "whenever our fiscal situation allows", he said.

He noted that Prime Minister Lee Hsien Loong said in his National Day Rally speech last August that $100 billion or more may be needed over 100 years to protect Singapore against rising sea levels caused by climate change.

Source: The Straits Time, https://www.straitstimes.com/singapore/singapore-budget-2020-new-coastal-and-flood-protection-fund-to-protect-singapore-against

EDUCATION



RESEARCH FUNDING

Request for Proposal (RFP) for Coastal Protection and Flood Management Research Programme (CFRP) Applied Research (Singapore)

National Research Foundation, Singapore (NRF)

The **Request for Proposal (RFP) for Coastal Protection and Flood Management Research Programme (CFRP) Applied Research** from the *National Research Foundation (NRF) of Singapore* supports research projects to adapt technology from other sectors in order to develop solutions for coastal protection and flood management. Applicants must be from Singapore-based Institutions of Higher Learning (IHLs), public sector entities and private sector companies. Grant applicants are strongly encouraged to collaborate with industry partners and overseas Institutes of Higher Learning. The R&D areas covered under this RFP include:

- Focus Area 1: To develop new sustainable construction materials that could enhance performance of Singapore's coastal protection structures.
- Focus Area 2: To establish proof-of-concept for Smart Structural Health Monitoring (SHM) System to monitor structural integrity of coastal infrastructures.
- Focus Area 3: To develop methodologies to support Integrated and Adaptive Planning for Coastal Protection
- Focus Area 4: To develop innovative adaptive coastal protection and flood protection solutions suitable for Singapore's context.
- Focus Area 5: To establish proof-of-concept for remote sensors or monitoring techniques for Singapore's sediment transport.

Source:
https://www.scientifyresearch.org/grant/coastal-protection-singapore-cfrp-ar/



Coastal Protection and Flood Resilience Institute Singapore

The Coastal Protection and Flood Resilience Institute (CFI) Singapore is the nation's first Centre of Excellence dedicated to strengthening local capabilities and expertise in coastal protection and flood management research and solution development.

CFI Singapore is a key pillar under PUB, Singapore's National Water Agency's S$125 million Coastal Protection and Flood Management Research Programme (CFRP), which galvanises research and technology development in coastal protection and flood management.

As the host of CFI Singapore, the National University of Singapore (NUS) works closely with Nanyang Technological University (NTU), Singapore University of Technology and Design (SUTD), Singapore Institute of Technology (SIT), and the Agency for Science, Technology and Research (A*STAR) to achieve the institute's key missions:

- Generate core knowledge in coastal science
- Conduct use-inspired research for coastal protection and flood management solutions for urban and land-scarce coastlines
- Collaborate to foster research ecosystem including local and international universities, research institutes and industry partners
- Train a new generation of leaders, researchers and practitioners for coastal protection and flood management

CFI Singapore's core and applied research is categorised into two horizontal domain areas (H1 and H2) and two vertical domain areas (V1 and V2).

NUS National University of Singapore | Coastal Protection and Flood Resilience Institute Singapore College of Design and Engineering | CFI SINGAPORE

Source:
https://cde.nus.edu.sg/cfisg/

BACKGROUND

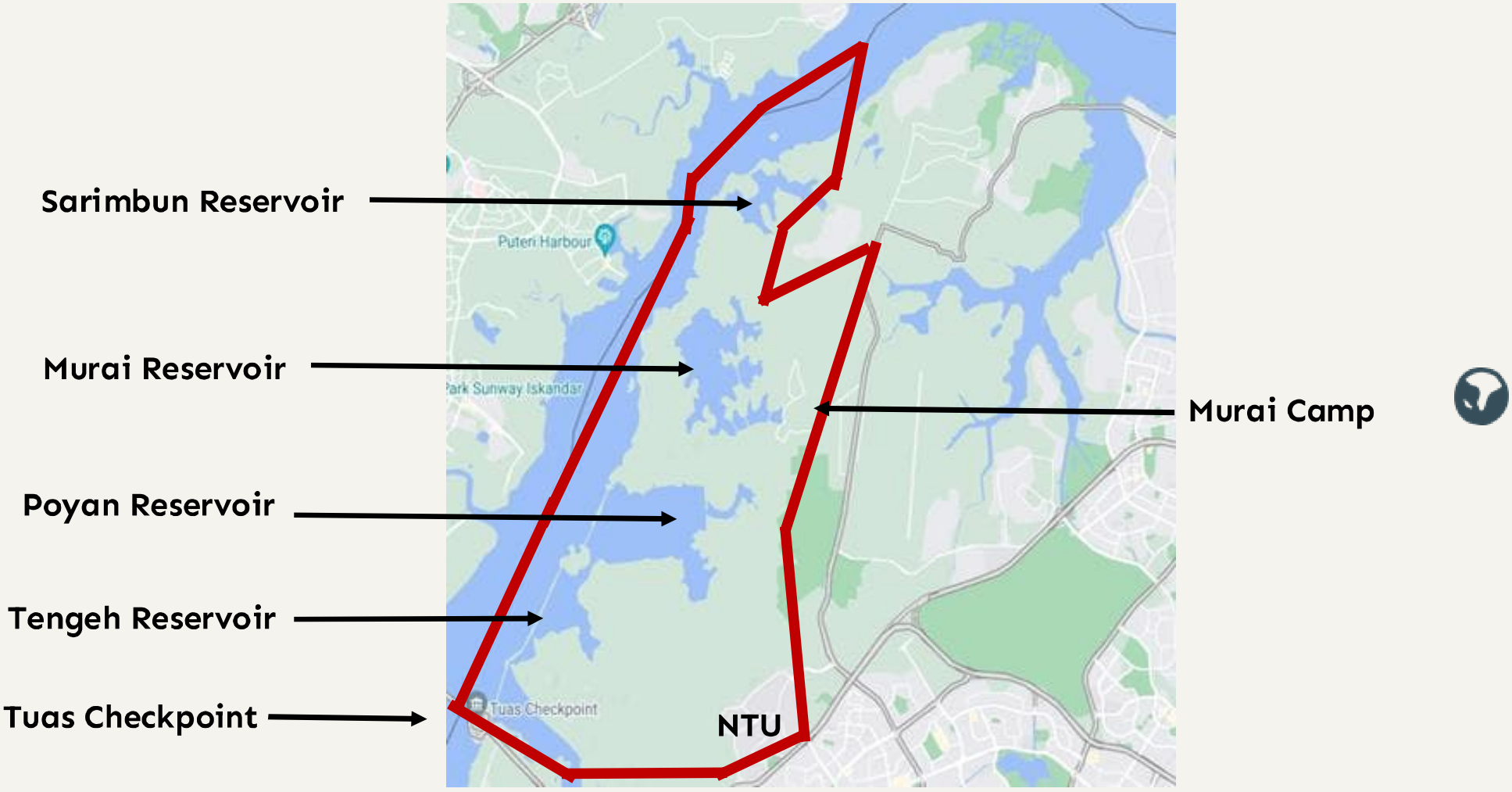




SITE ANALYSIS

02

Northwest Coast 1 Site Analysis (C1)



Overview of Northwest Coast 1 (C1)

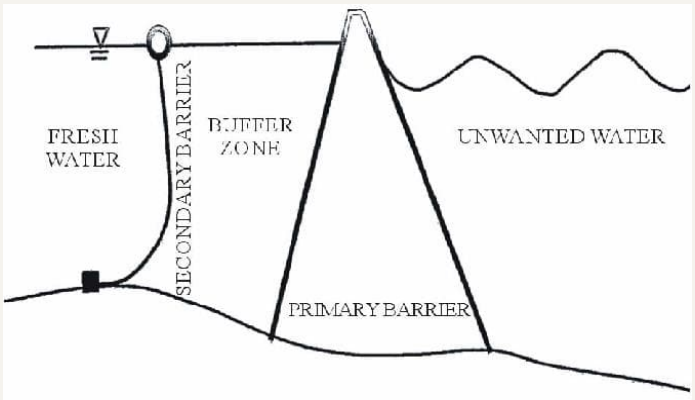
Water Supply and Water Catchment

- 4 coastal reservoirs
- Issues that threatening the drinking water supply:
  - Sea Water Intrusion
  - Wave Action
  - Sea Level Rising

Potential issues? or it is already issues? If this is existing problem, please cite source (media, etc.)



Source: www.timeout.com



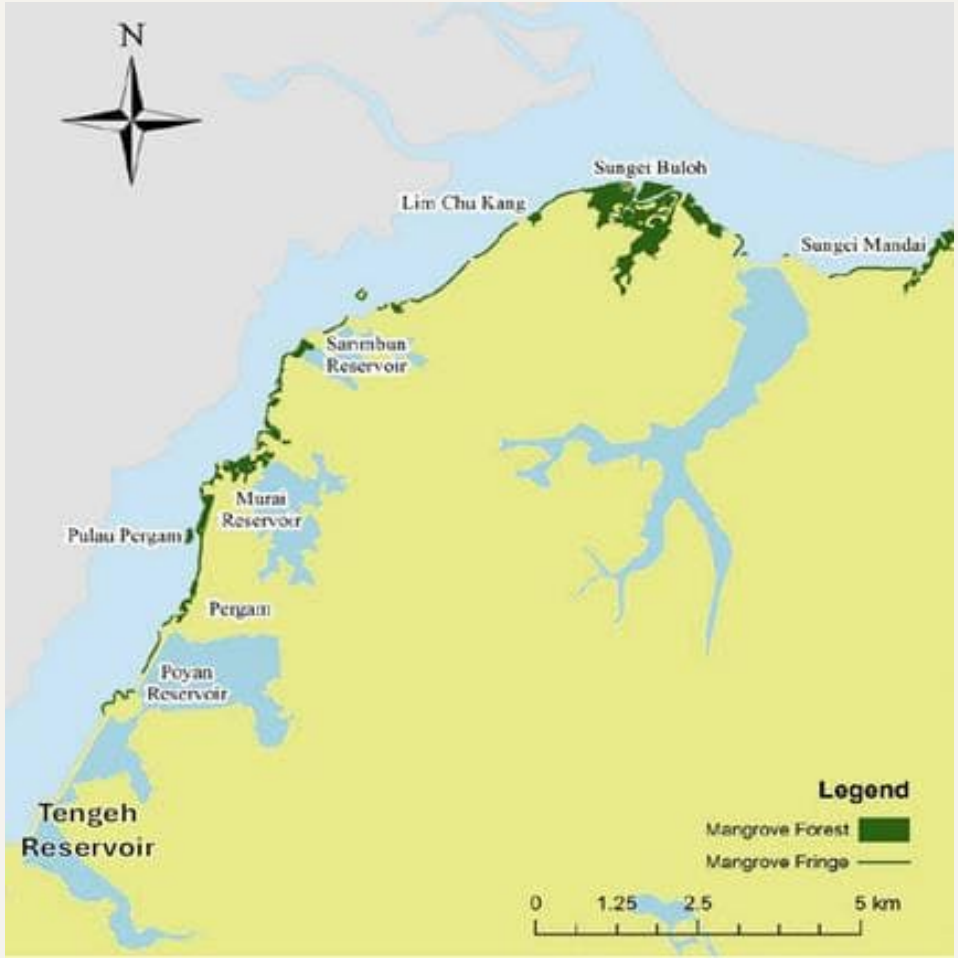
Source: www.researchgate.net

Overview of Northwest Coast 1 (C1)

Biodiversity

- Mangrove forests along the coastline
- Mangrove family: Rhizophoraceae, Avicennia, Lumnitzera
- Habitat for various marine and terrestrial organism
- Seagrass was common between 1950s - 1970s

highlight issue here, e.g. erosion which causes mangrove population to decline, etc.



Source: https://www.researchgate.net/publication/268374524_The_present_extent_of_mangrove_forests_in_Singapore

Overview of Northwest Coast 1 (C2)

Tengeh Solar Farm & Tuas Checkpoint

- Largest floating solar farm
- Clean energy
- Power supply to 16,000 home
- Ambitious to invest in more and mature floating solar panel
- Border issues

to elaborate if possible



Source: mothership.sg



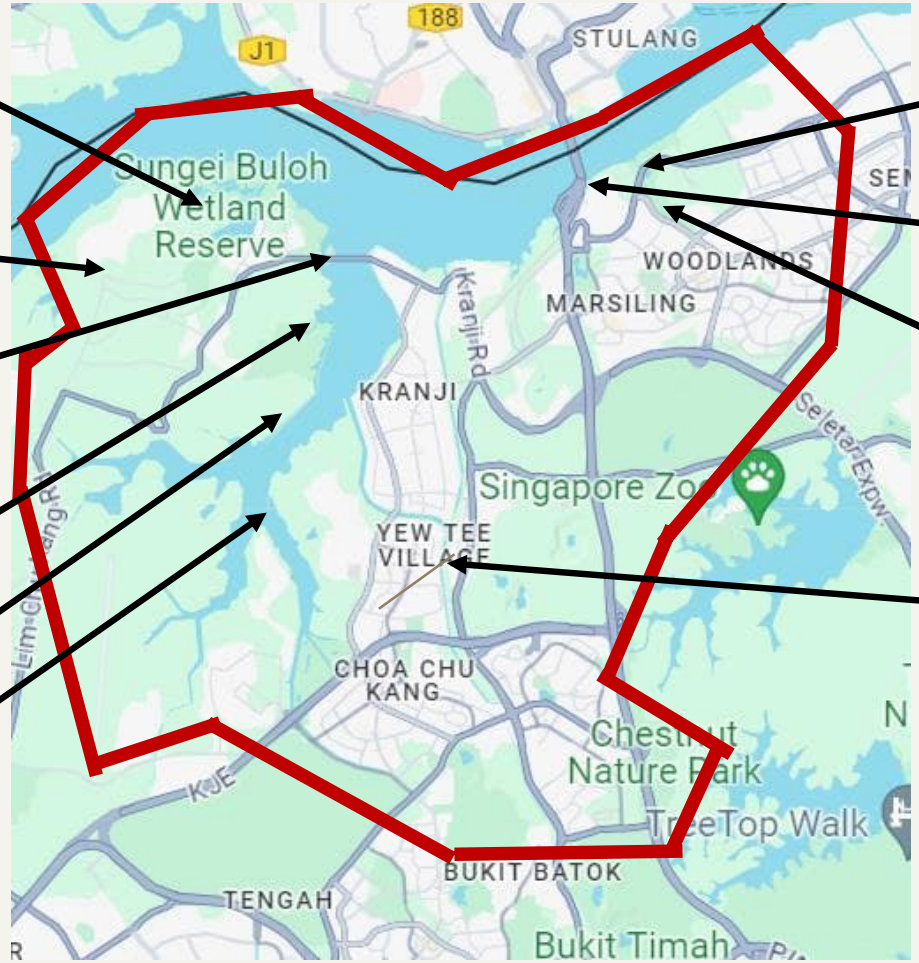
Source: m.malaysiakini.com





Source: www.straitstimes.com

Northwest Coast 2 Site Analysis (C2)



Sungei Buloh Wetland Reserve

Lim Chu Kang Nature Park

Kranji Dam

Urban Agriculture District

Kranji Marshes

Kranji Reservoir

Woodlands Jetty

Woodlands Checkpoint

Woodlands Waterfront Park

Sungei Kadut Eco-District

SUNGEI BULOH NATURE PARK NETWORK

FINANCIAL

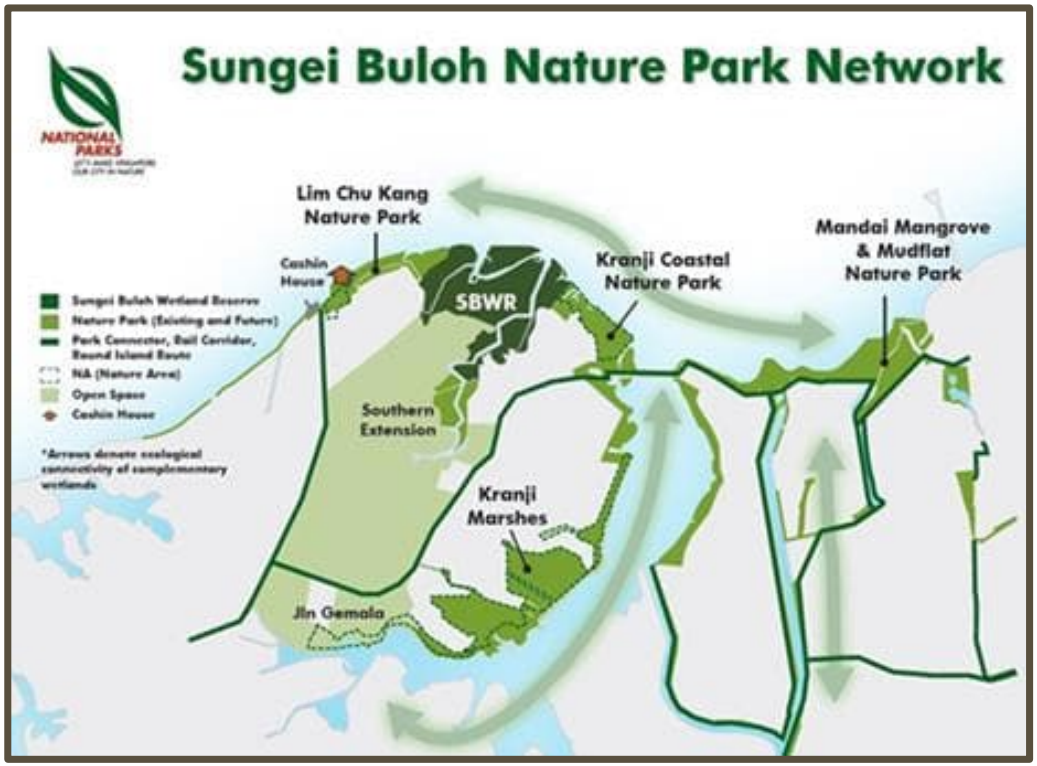
- Contributor to nature parks' income
- Recreational & tourism attraction
- Natural barrier for Kraji Reservoir

typo

ENVIRONMENT

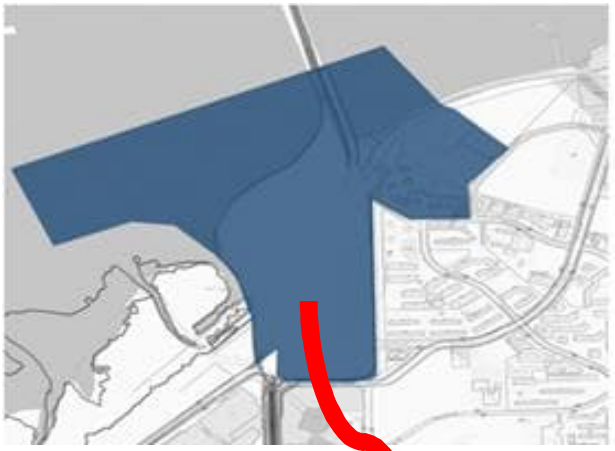
- **500+ species of flora and fauna**
- Mangroves act as buffer line against sea level rise & erosion

any issue here? if yes, good to highlight



Source: *beta.nparks.gov.sg*

Future Development





Woodlands Checkpoint Expansion

THE STRAITS TIMES SINGAPORE

Lim Chu Kang set to be redeveloped into high-tech agri-food cluster: SFA



Lim Chu Kang Urban Agriculture District

Source: https://str.sg/JBHn

Sungei Kadut Eco District

Source: https://str.sg/JpsU

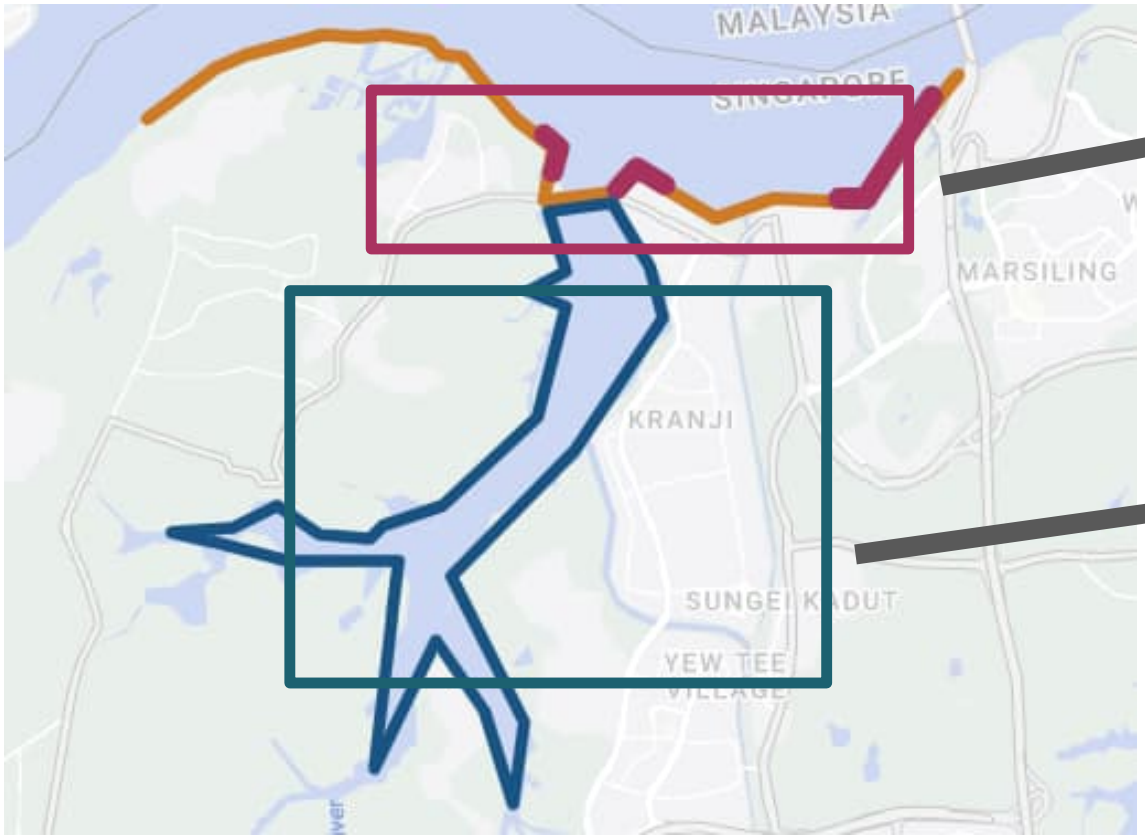
THE STRAITS TIMES

High-rise industrial facilities, agri-tech park feature in revamped Sungei Kadut industrial estate





Erosion and Salinization Degree along C2



Medium to High Level of Erosion C2

High Degree of Salinization Vulnerability Degree

any proof? media coverage? good to cite, if yes.

Source: https://www.nparks.gov.sg/-/media/cuge/ebook/citygreen/cg19/full/cg19_03.ashx

what kind of impermeable surface? you mean like buildings and infrastructures built on the reclaimed land?

Urbanisation: The Issues Behind

Flooding



- Land reclamation alters natural water flow
- High-tech farming may have alterations to land and drainage systems

Erosion



- Reclamation disturbs natural sediment balance
- Removal of vegetation causes erosion and sediment loss

Runoff



- Impermeable surface pollutes water & causes flooding
- Agrochemicals in farming can lead to contaminated runoff



SOLUTION

03

PRINCIPLE OF SOLUTIONS

FEASIBILITY



SUSTAINABILITY



COST-EFFECTIVENESS



INNOVATION



Xbloc Revetment with Ecological Enhancements

Benefits

High stability:

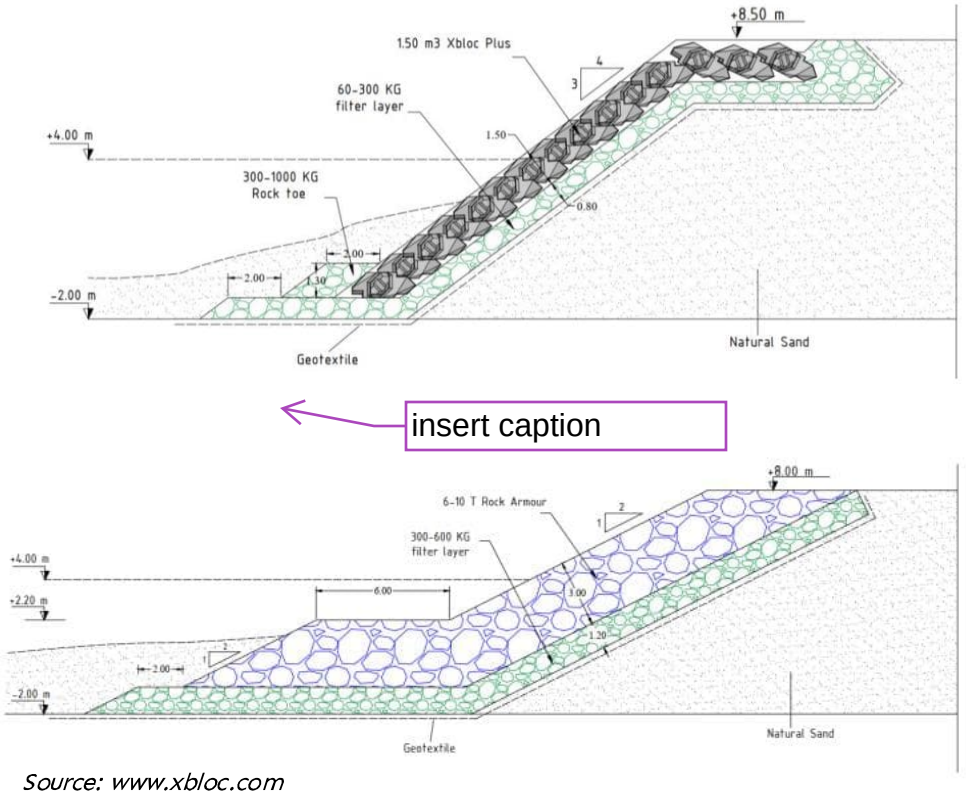
- Scour prevention (Toe structure)
- Wave energy Dissipation (Up to 30 feet)

Material efficiency:

- Reduced armor layer thickness (40% less)
- Durability (100 years design life)

Environmental-friendly:

- Wider shoreline (3:4 slope)
- CO_2 reduction (10-50% less)



Source: www.xbloc.com

insert caption

insert caption

i think you can say more usable land area because steeper slope (3:4) can be adopted instead of conventional 1:2

from the top layer only, right? as the bottom layer you still need some class B/C stones - can elaborate when presenting

can elaborate from where is the reduction

Xbloc Revetment with Ecological Enhancements

Xbloc as rock armour unit

- Interlocking design allow easy installation
- Less labour, riggers for lifting is not required

  (Using hydraulic gripper)
- Recycled material
- Maintenance free in 50 years-lifetime
- Hollow design conserve marine life
- Against wave up to 200% designed height







design life is 100 years, maintenance free for 50 years. what to do after 50 years? can elaborate when presenting

Source: www.xbloc.com

Xbloc Revetment with Ecological Enhancements

Ecological Enhancements

Coastal plants:

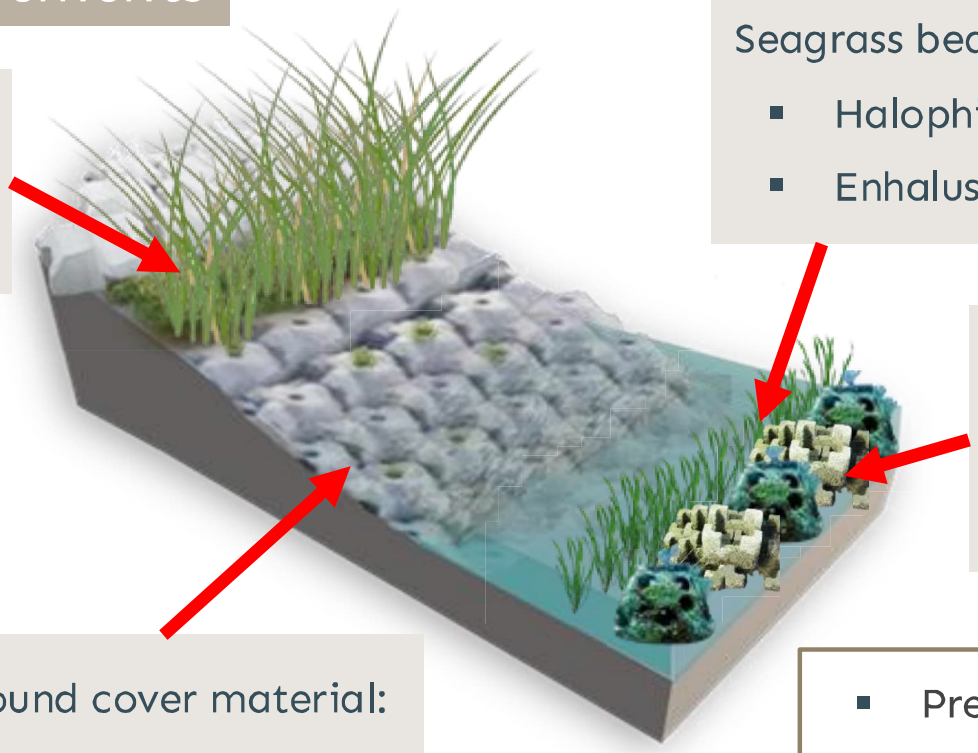
- Sporobolus virginicus
- Sesuvium portulacastrum



Source: www.globalsynthetics.co.nz



Source: SlingBag.net



Seagrass beds:

- Halophila beccarii
- Enhalus acoroides

Artificial reefs:

- Reef ball
- Oyster castle

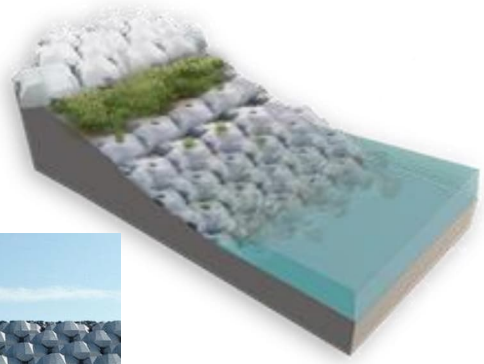
Ground cover material:

- Geotextile mat
- Burlap sacking

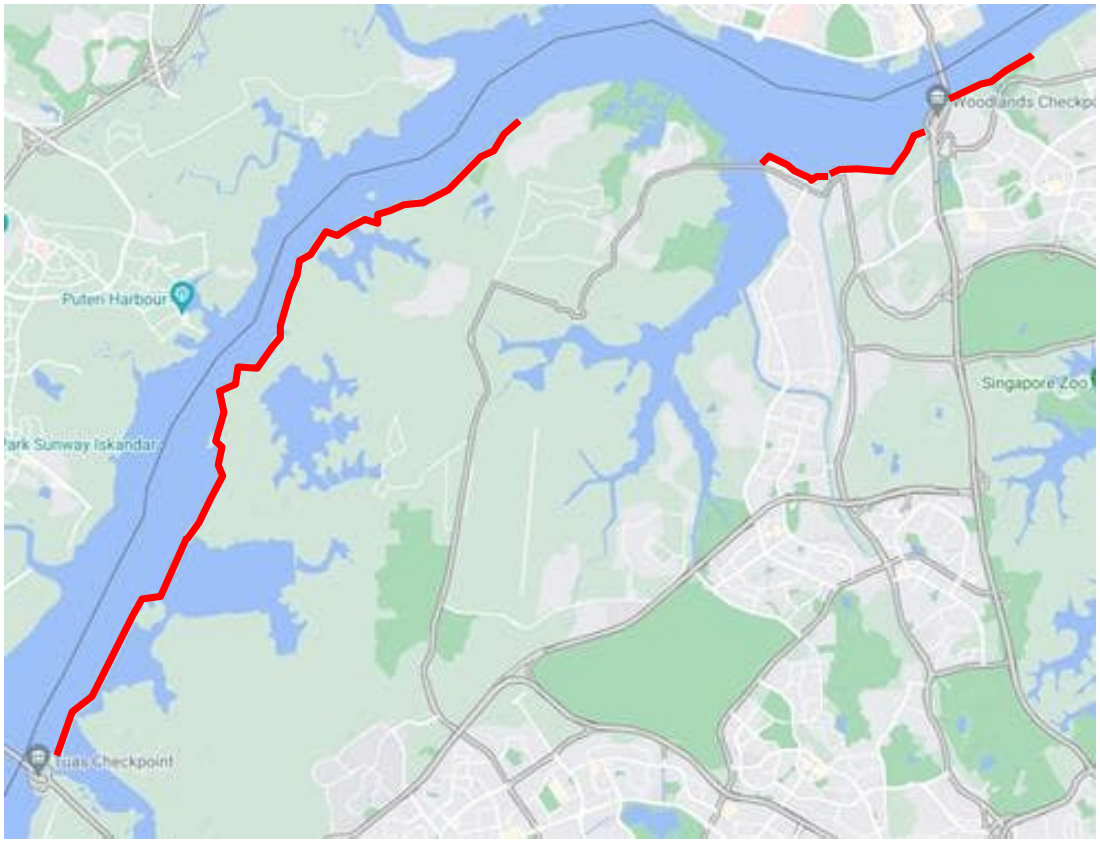
- Prevent seawater intrusion
- Reinforced revetment
- Restored biodiversity

Solution Overview

Xbloc Revetments with Ecological Enhancements



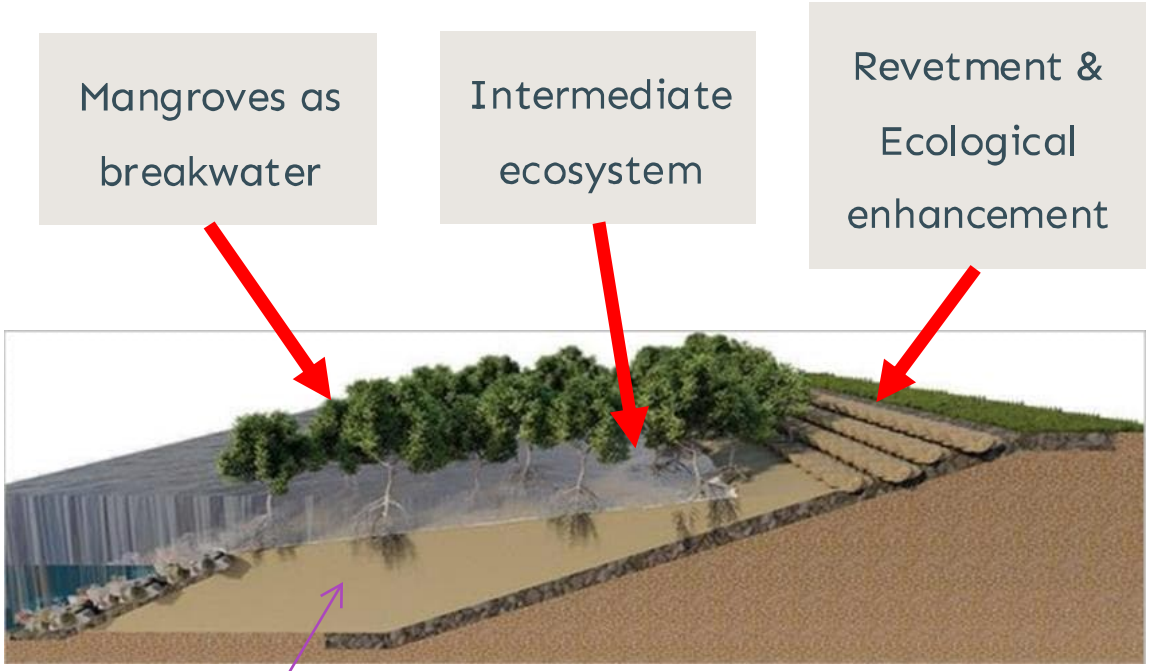




Mangrove Expansion utilising tetraPOT

Functions of Mangroves

- Nature's best breakwater
- Desalination
- Hybrid protection system
- Promotes ecosystem growth



Source: https://www.researchgate.net/figure/Hybrid-coastal-protection-system-consisting-of-nature-based-feature-eg-mangrove-and_fig1_364198836

note that mangrove needs to be in intertidal zone

Mangrove Expansion utilising tetraPOT

TetraPOT Design

- Dimensions: 1.5m x 1.5m x 1.5m
- Material : Sustainable Concrete
- Manufacture: Prefabrication
- Can be adjusted to fit required depths
- Ease of manufacture & plantation

  (reduced in-situ works)



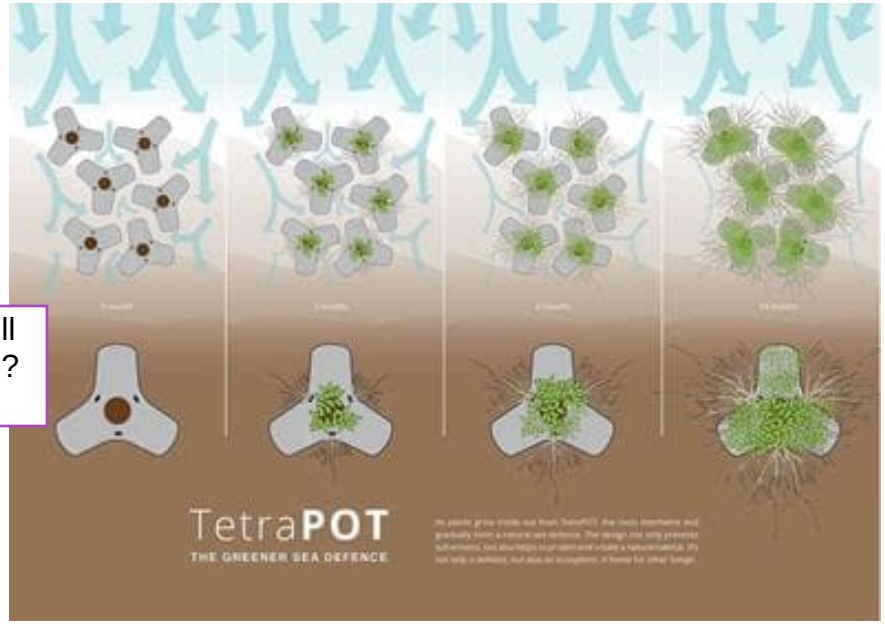
Source: https://www.shenghunglee.com/tetrapot

Mangrove Expansion utilising tetraPOT

How it works

- Mangroves helps lock the tetraPOTs in place
- Approximate time for Mangroves to be in effect: **1 year**
- Benefits of hard engineering while maintaining natural landscape

sure it is 1 year for it to grow to full size and act as coastal protection? seems a bit short to me...



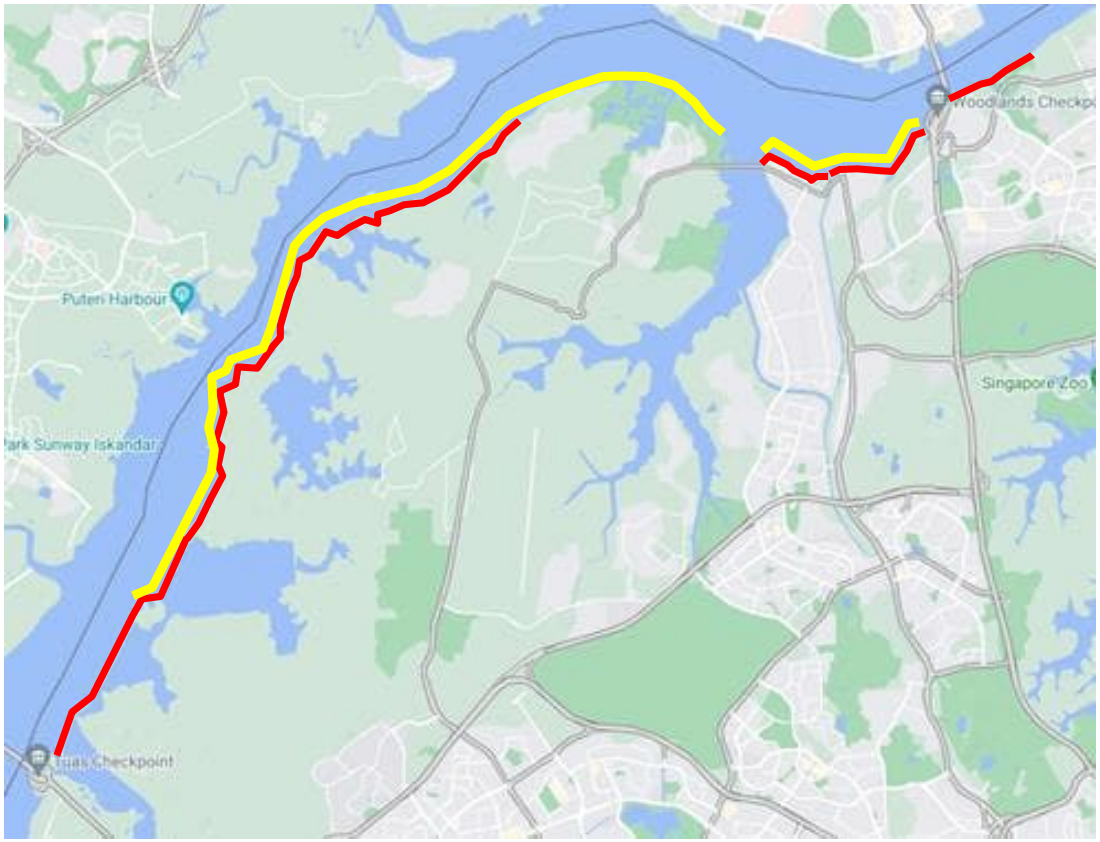
Source: https://www.shenghunglee.com/tetrapot

can add how long the life span of a tetrapot. will it continue to house the mangrove? or will it break and disintegrate after sometime allowing the mangrove to root in the seabed?

Solution Overview

Mangrove Expansion





Solar Panel Integrated Floating Breakwater

Benefits

- Expansion to current Tengeh solar farm
- Rather calm waves in West Johor Strait
- Upgraded design to face sea conditions
- Compared to conventional breakwaters:
  - Less concerns for border issues
  - Less harmful to marine life

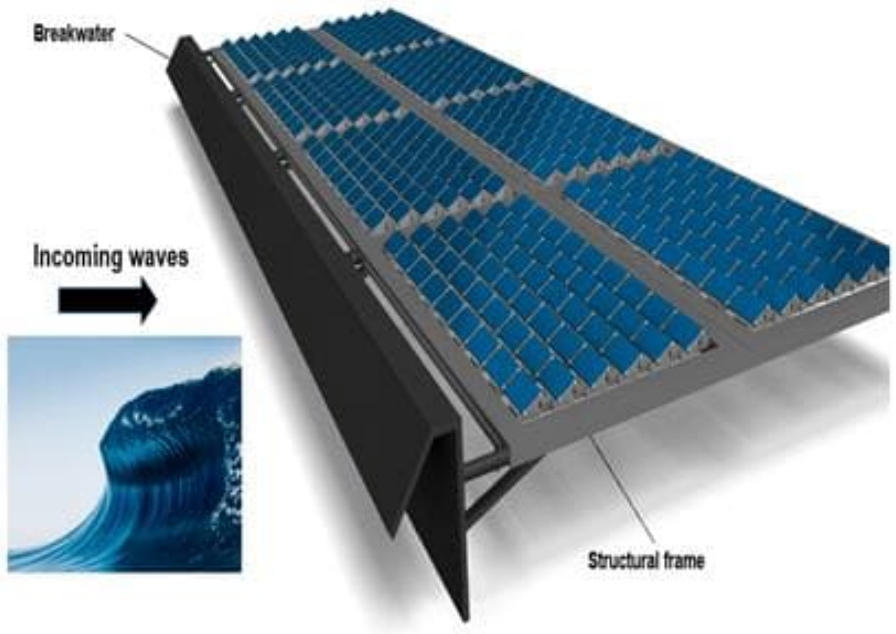


Image for illustration purposes only

Source: https://www.mdpi.com/2071-1050/16/3/1345

this one i am not so sure, must make sure you are not encroaching into singapore/malaysia border or the navigational channel

Solar Panel Integrated Floating Breakwater

Feasibility

elaborate

- Ease of Maintenance
- Modular design (replaceable parts)
- Monitoring system

such as?

- Anchored to seabed
- Use area as testbed for more innovative designs

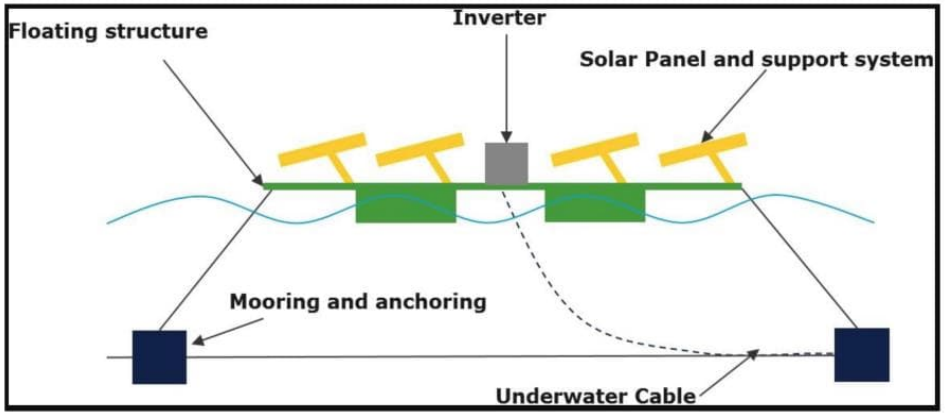


Image for illustration purposes only

Source: https://www.mdpi.com/2571-8797/4/3/46

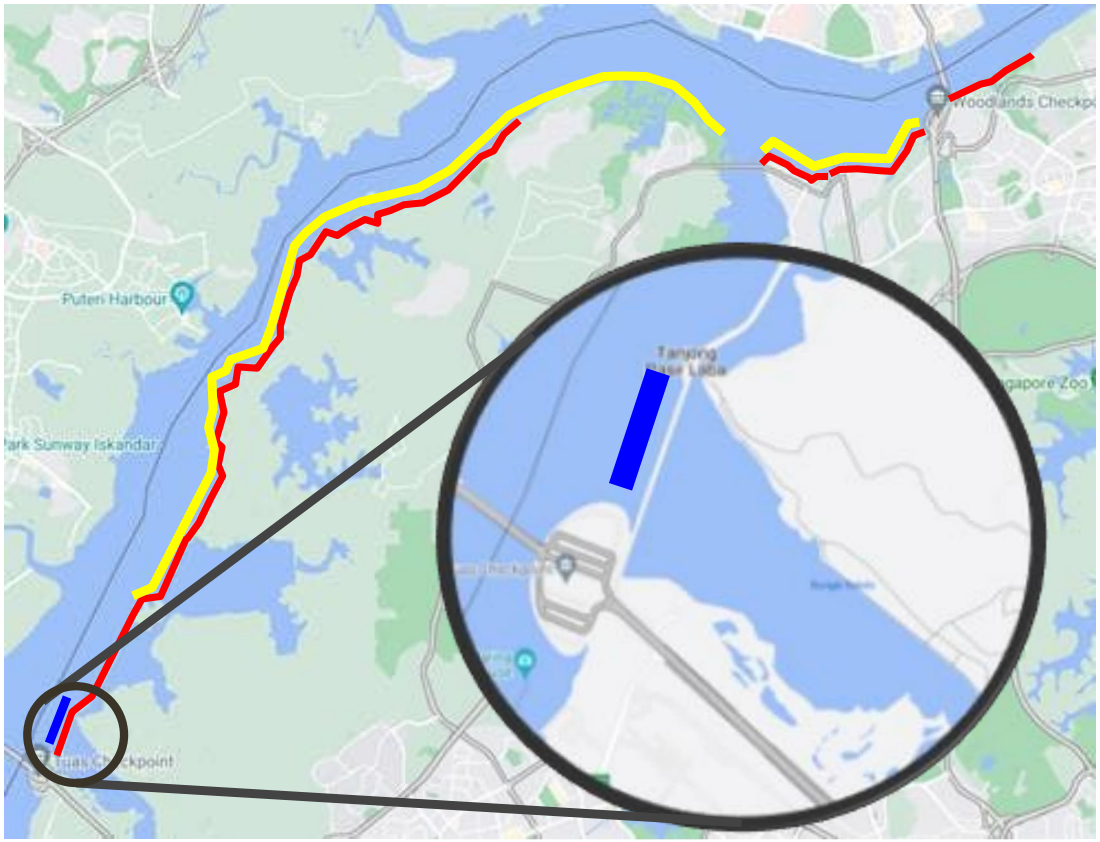
i highlighted corrosion issue with the material that you selected, but it's still not addressed?

Solution Overview

Solar panel integrated floating breakwater



Source: https://www.en-former.com/en/how-to-build-a-floating-solar-farm/

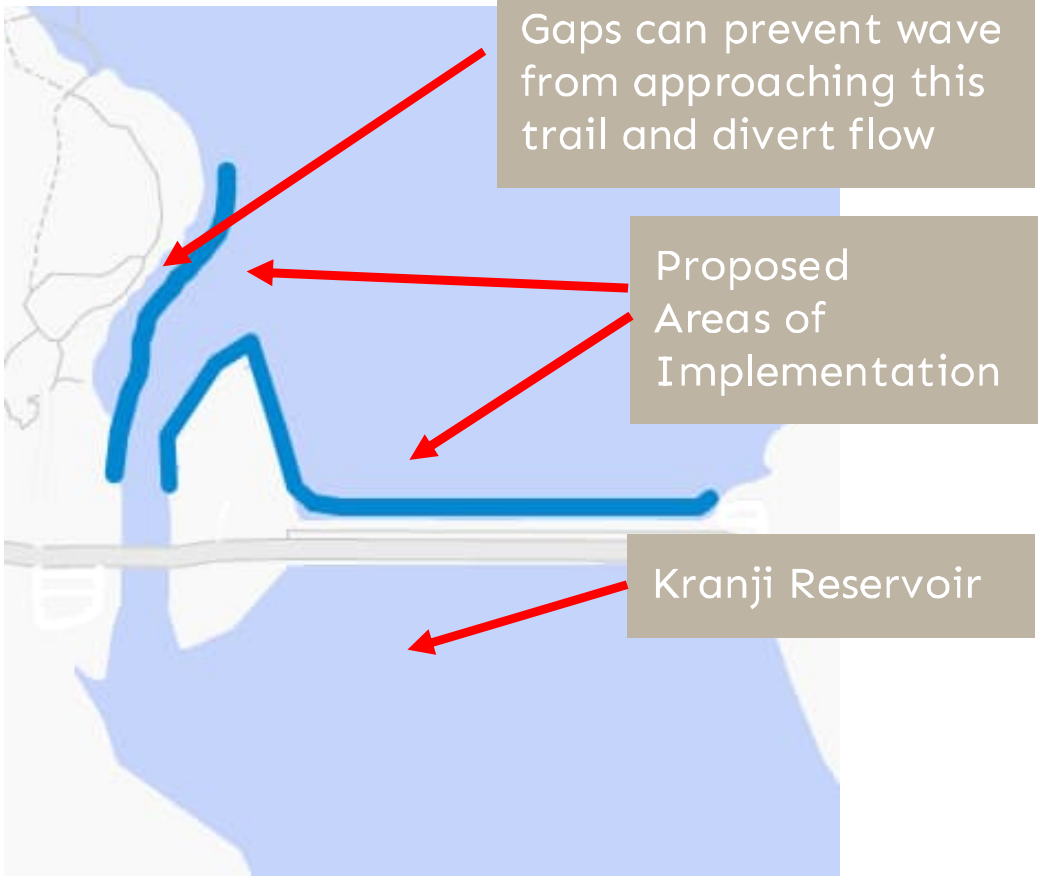


Flexible Rubber Tyre High-Strength Net Breakwater



Functions

- Reduce wave energy
- Help sediment deposition
- Increase roughness
- Resilient to changing wave conditions and adaptable to coastal environments.



Source: https://jmstt.ntou.edu.tw/cgi/viewcontent.cgi?article=1483&context=journal

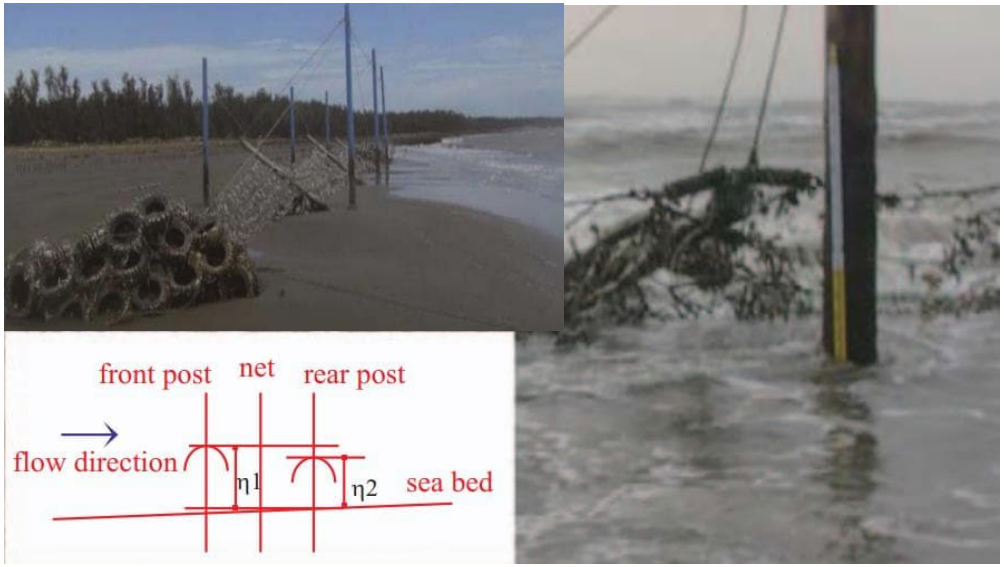
can highlight the height you intend to install the tyre breakwater, it shd be higher than the expected wave etc

how does this work? just stack tyre to form a bund? or inside you need some sand? if there is any cross section view it will be good

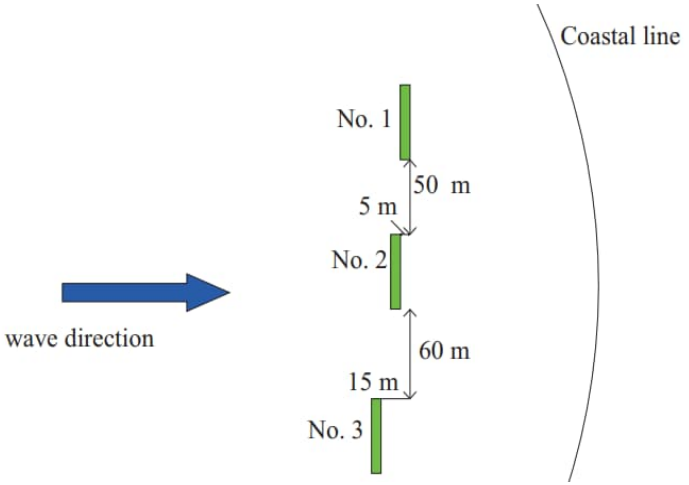
Flexible Rubber Tyre High-Strength Net Breakwater

Benefits

- Lower capital cost
- Quick and Simple to Implement
- Non-toxic
- Ecologically Safe with Tidal Impedance



Source: https://jmstt.ntou.edu.tw/cgi/viewcontent.cgi?article=1483&context=joumal



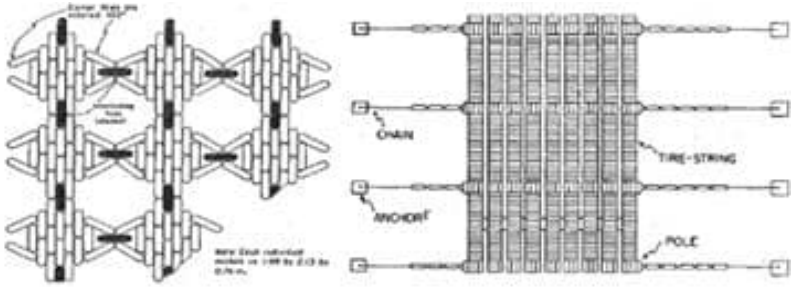
Source: https://jmstt.ntou.edu.tw/cgi/viewcontent.cgi?article=1483&context=joumal

Flexible Rubber Tyre High-Strength Net Breakwater

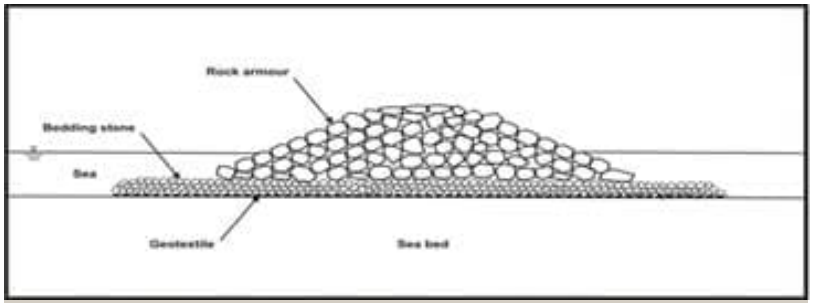
Comparison with Conventional Breakwater

- Cost effective, materials readily available.
- Flexible, dissipate wave energy more effectively. (adapt to varying wave conditions and water levels)
- Promotes sediment deposition, creates habitats and supports biodiversity.
- Recycled materials support circular economy.

maintenance?



Flexible High-strength Rubber Tyre Breakwaters



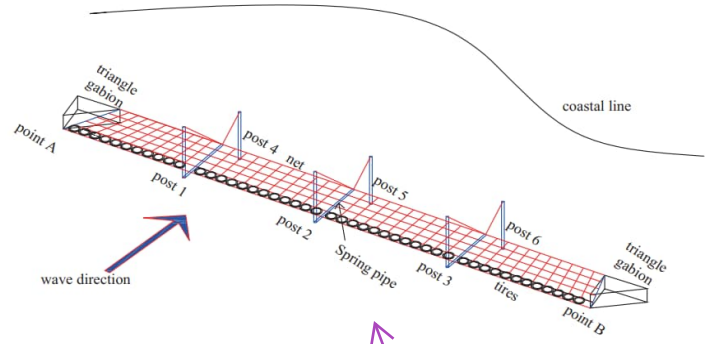
Rubber Mound Breakwaters

rubble

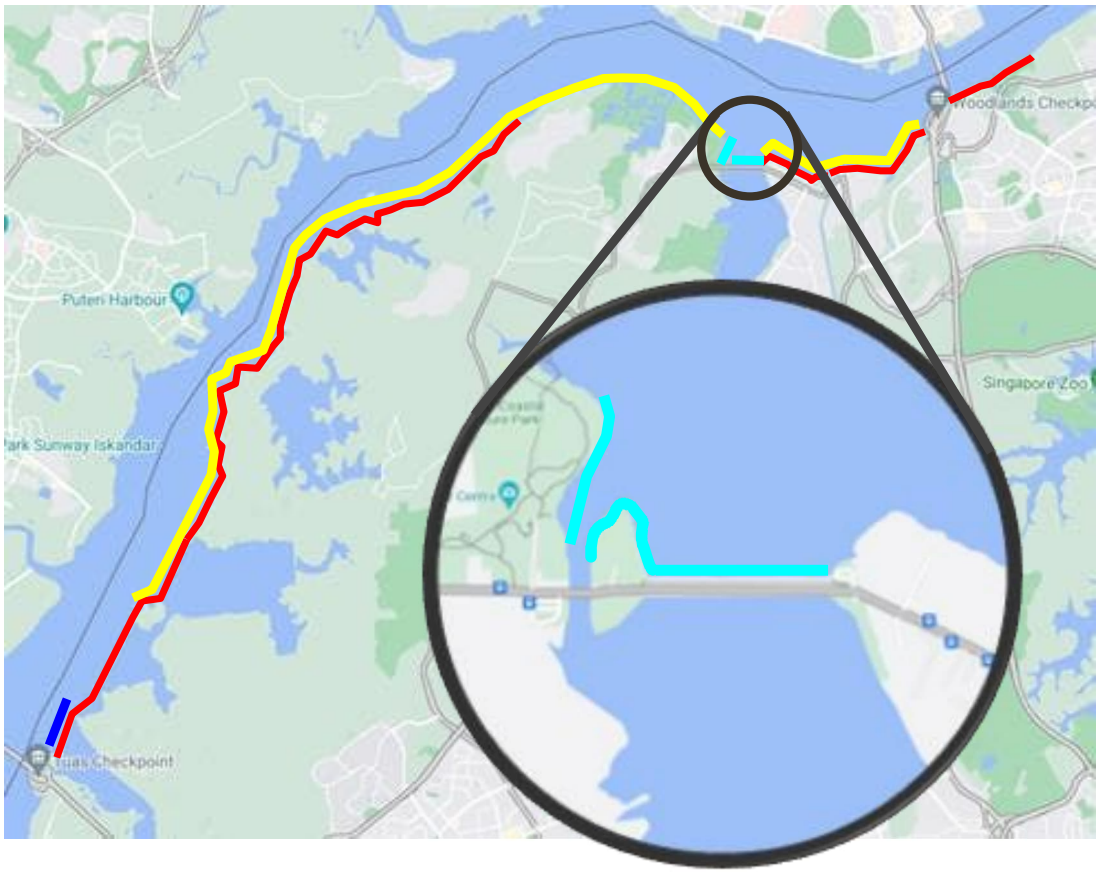
Solution Overview

Flexible Rubber Tyre High-Strength Net Breakwater





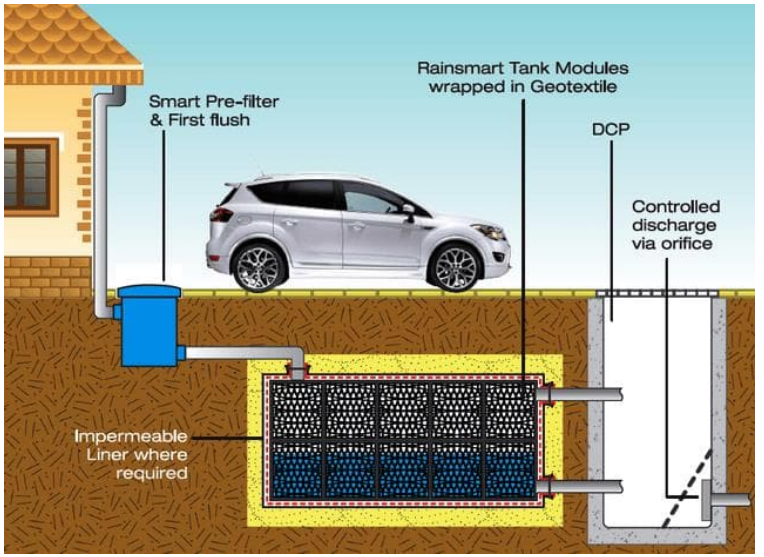
what is this? different with what you show in the previous slides



Modular Detention Tanks

Functions of MDTs

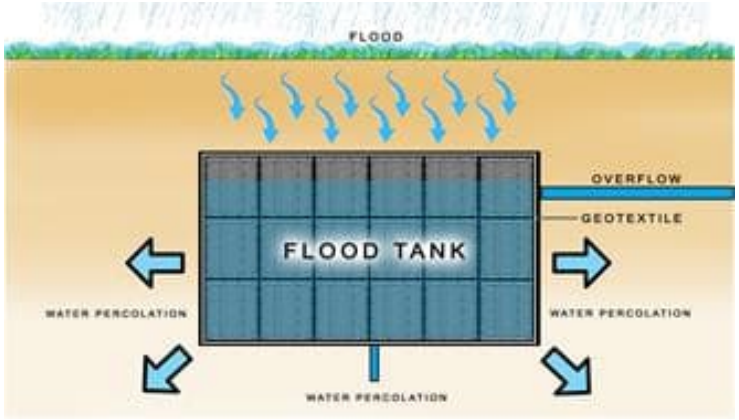
- Reduce risk of flooding
- Peak discharge rate reduction
- Integration with existing drainage
- Stormwater storage & reuse



Source: *www.rainsmartsolutions.com*



Source: *stormwater.brentwoodindustries.com*

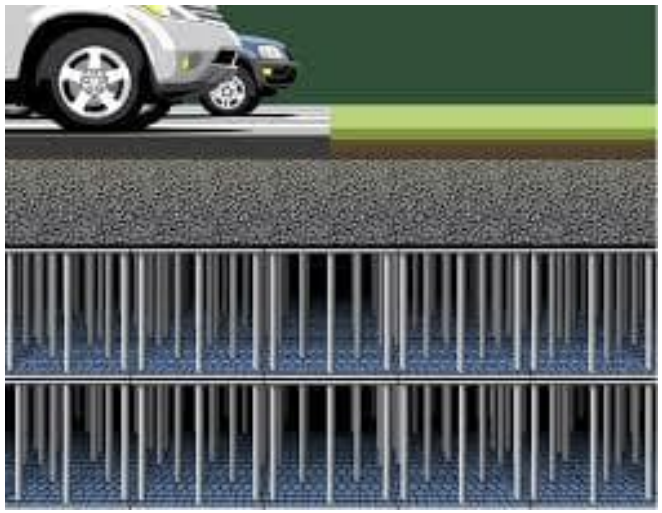


Source: *vertigrow.filipinasplastic.com*

Modular Detention Tanks

Practicability

- Suitable for both pervious & impervious surfaces (e.g. vegetation & concrete)
- Handle 60 Tons per square meter
- Minimum 60 years design life
- Implement during reclamation help to reduce cost for excavation
- Scalable to Future Sungei Kadut Eco District for non-potable use



Source: stormwater.brentwoodindustries.com

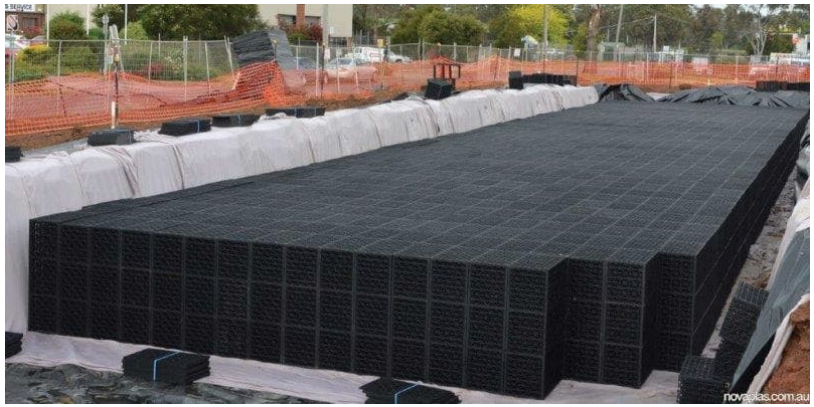


in reality a bit difficult especially when you dont know yet what are the exact things that you will build on the reclaimed land. but if during the planning stage, the masterplan has been developed, the tank can be built as reclamation progresses (might still need excavation tho)

Modular Detention Tanks

Comparison with Concrete Detention Tanks

- Subsurface infiltration
- Higher void space ratio (95%)
- Flexible & light weight
- Recycled materials
- Easy to install (1 labour = 75 m^3 per day)



Source: *novaplas.com.au*

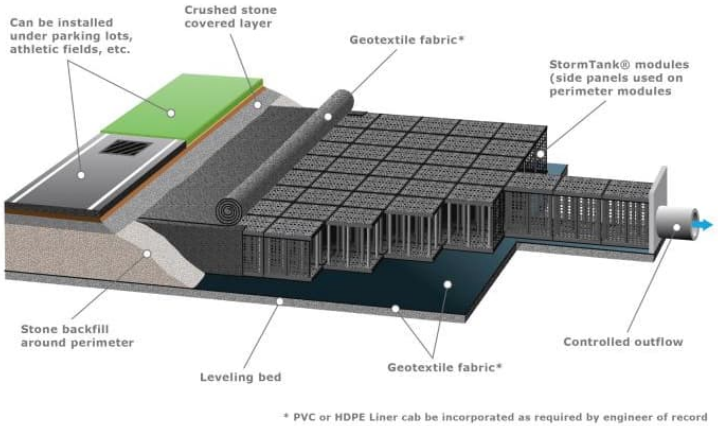


Source: *atlan.com.au*

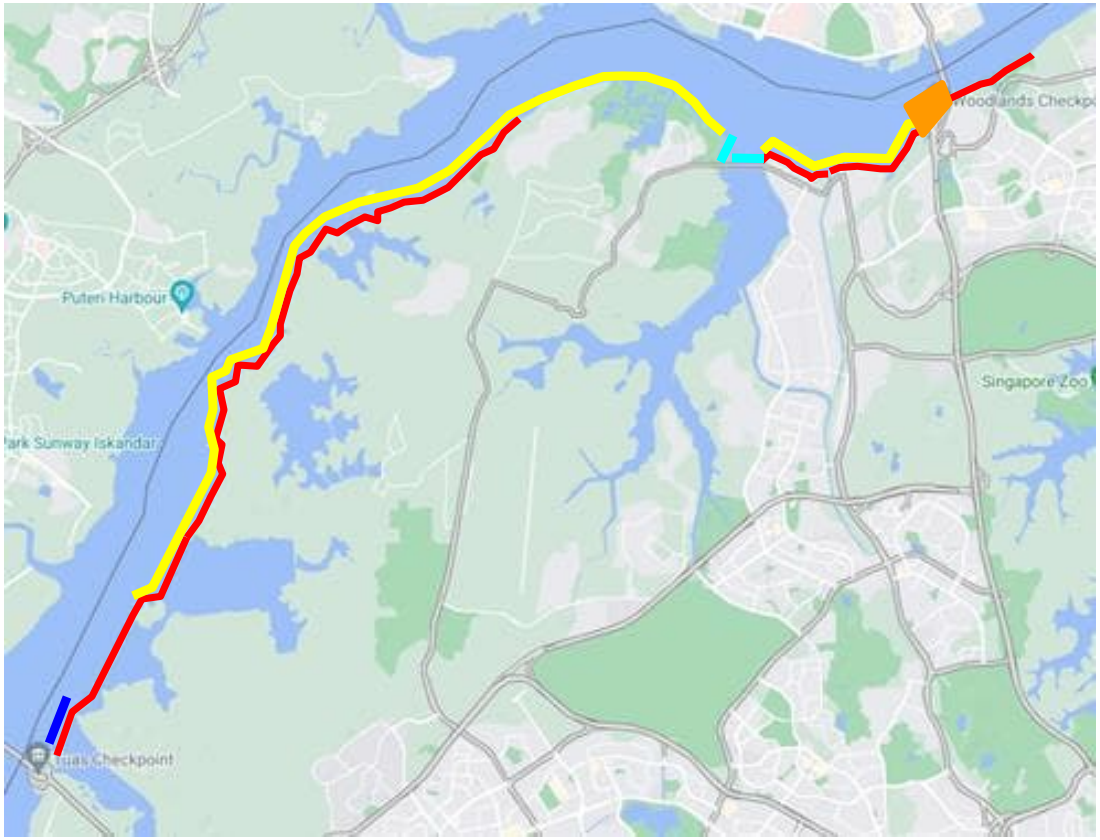
Solution Overview

Modular Detention Tanks





* PVC or HDPE Liner cab be incorporated as required by engineer of record



Revamp of Woodlands Jetty

Importance

- Historical Heritage
- Recreational Activities



Negative Impacts

- Pilings → Habitat destruction
- Shading → vegetation decline



Source: beta.nparks.gov.sg

Revamp of Woodlands Jetty

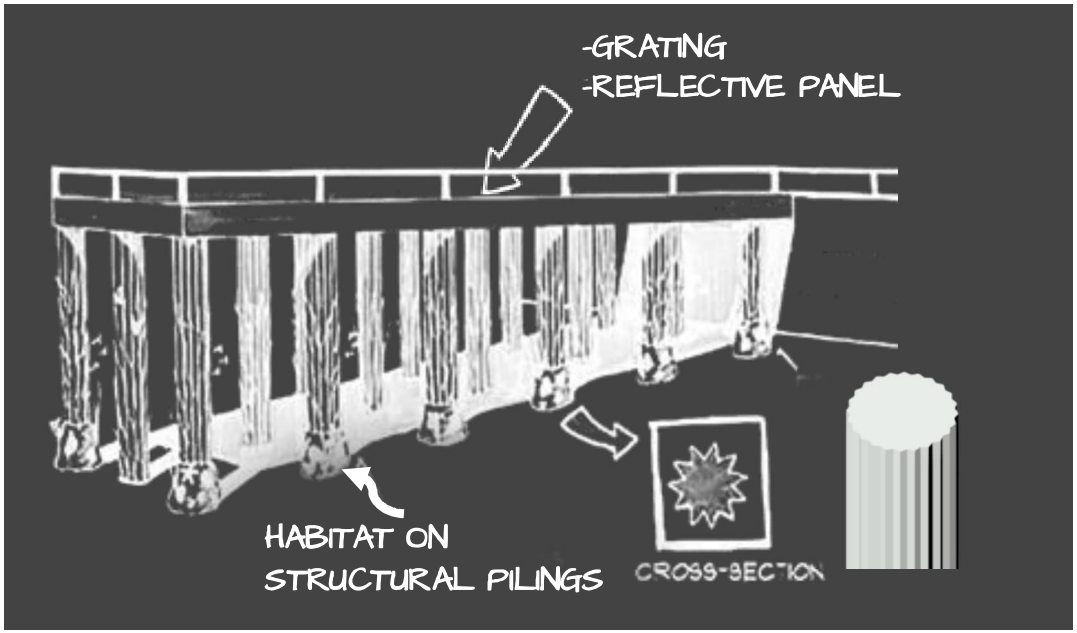
Design details

Microhabitats:

- Textured pilings & Fish shelters

Lighting:

- Fiberglass grating & Reflective panel



Main goal:

Restore biodiversity loss caused by

Woodlands Checkpoint expansion



Source: www.scmp.com



Source: www.archatrak.com

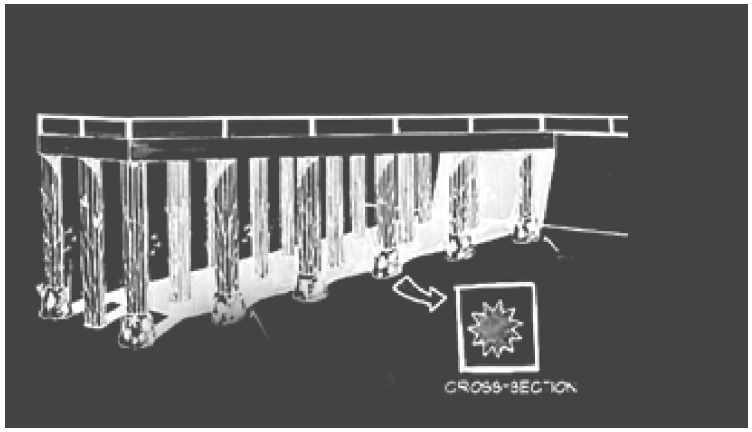


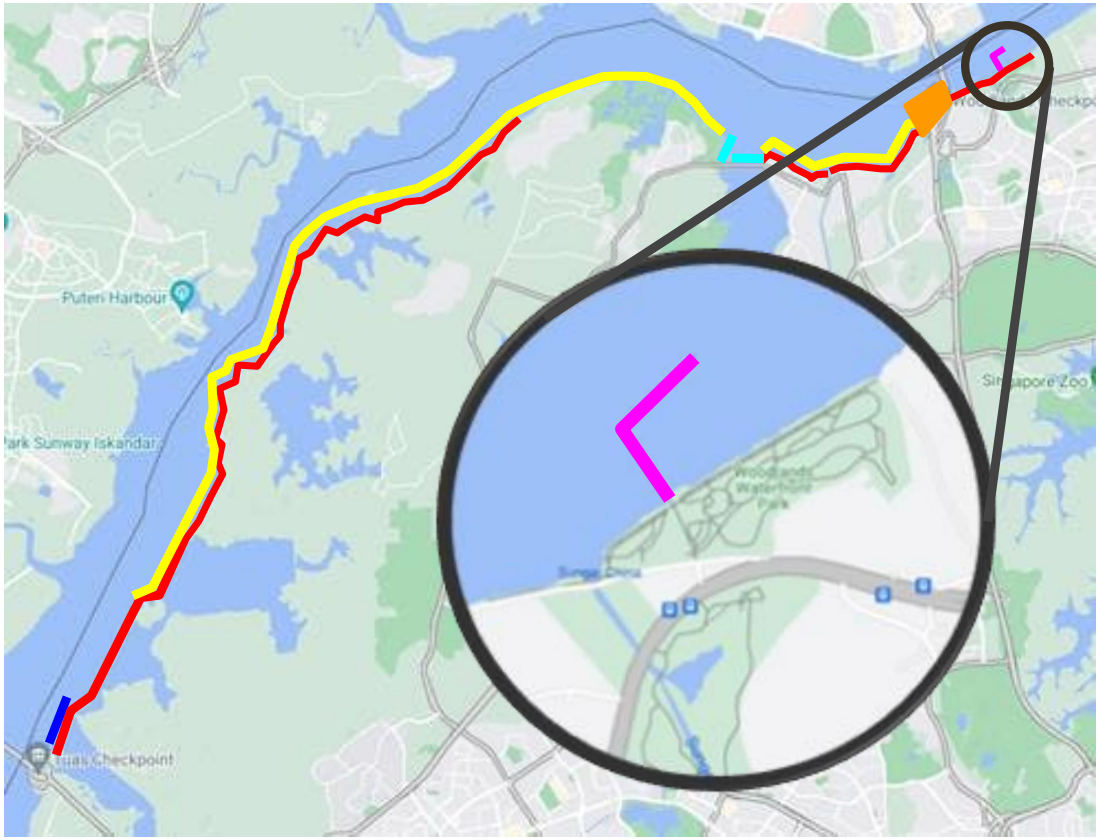
Source: conservation.reefcause.com

Solution Overview

Ecological Jetty









ASSESSMENT

04

Cost Consideration



Initial Cost

- Materials
- Site survey
- Consultation
- Engineering & design
- Equipment & machinery



Implementation Cost

- Integration
- Contingency
- Transportation
- Testing & monitoring
- Construction & installation



Maintenance Cost

- Utilities
- Repairment
- Operational
- Routine inspections
- Environment assessment

Cost Category

		Cheap	Average	Expensive
Initial Cost	Per m^3	100	1k	
	Per m^2	1k	5k	
	Per m	5k	10k	
Implementation Cost	Per ha(10000m^2)	500k	2.5Mil	
Maintenance Cost	Percentage of initial + implementation cost (Annual)	1%	5%	

Cost Analysis

Xbloc Revetment with Ecological Enhancement

Scale: 21km coastline

Time span: 3.5 years

Initial: SG$5333/m

Implementation: 1.36Mil/ha

Maintenance : 0.8%/year

Mangrove Expansion utilizing tetraPOT

Scale: 30ha

Time span: 2 years

Initial: SG$67/m2

Implementation: 467k/ha

Maintenance: 2.6%/year

Modular Detention Tanks

Scale: 250,000m3

Time span: 1 year

Initial: SG$98/m3

Implementation: 1.02Mil/ha

Maintenance: 0.9%/year

Solar panel integrated Floating Breakwater

Scale: 6ha

Time span: 1 year

Initial: SG$417/m2

Implementation: 733k/ha

Maintenance: 2.7%/year

Flexible Rubber Tyre High Strength Net Breakwater

Scale: 1.2km coastline

Time span: 6 months

Initial: SG$483/m

Implementation: 417k/ha

Maintenance: 4.6%/year

Revamp of Woodlands Jetty

Scale: 400m renovation

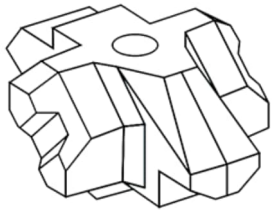
Time span: 4 months

Initial: SG$4325/m

Implementation: 1.6Mil/ha

Maintenance: 3.4%/year

Total Cost Overview



Total cost: 275.5 Million



Total cost: 43 Million



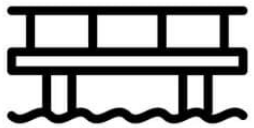
Total cost: 37.7 Million



Total cost: 37.4 Million



Total cost: 1.58 Million

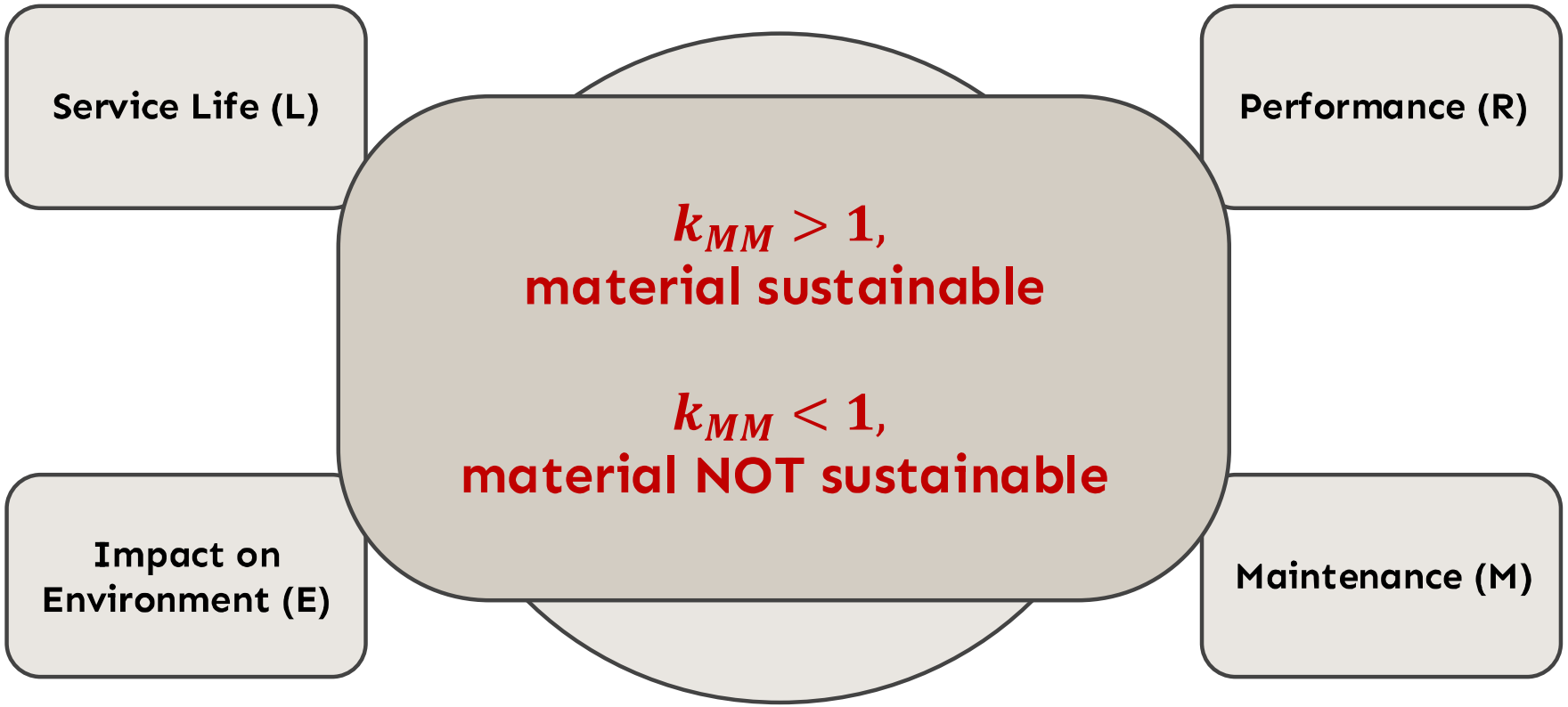


Total cost: 2.93 Million

Overall Project Cost Estimation: SG$ 398.11 Million

i am not savvy with costing, but i hv a feeling this is on the lower spectrum of the cost, considering area that you cover and so many things that you intend to do

Sustainability Coefficient



** X_{ref} = Reference value of X

Sustainability Quantification Example

RAC = Recycled Aggregate Concrete

Service Life (L)

Normal concrete: 100 years
RAC: 97 years

Service Life Ratio = 0.97

Performance (R)

Normal concrete: 52 MPa
RAC: 53 MPa

Performance Ratio = 1.019

Impact on Environment (E)

Eco Cost of Normal Concrete: 0.125 USD/Kg
Eco Cost of RAC: 0.11 USD/Kg

Environment Ratio = 0.88

Maintenance (M)

Both ordinary and sustainable product do not have significant difference

Maintenance Ratio = 1

Sustainability Coefficient = 1.123
(>1, Sustainable !)

Impact Analysis

Triple Bottom Lines



Economy

- Prevent Potential Economic Loss
- Support Circular Economy
- Cost Effective



Environment

- Biodiversity Preservation
- Climate Resilience
- Reduce Carbon Footprint



Social

- Community Safety
- Safeguarding Drinking Water Supply
- Recreation and Education Resources

Impact Assessment

Solution	Economic	Environmental	Social	Feasibility
Mangrove expansion with tetrapots				
Xbloc revetments with ecological enhancements				
High strength net tyre breakwater				
Solar panel integrated breakwater				
Ecological Jetty				
Modular Detention Tank				

OUR STORY ENDS HERE.

THANK YOU!

SPECIAL THANKS TO:

ARUP

BECC Built Environment Case Competition

PUB SINGAPORE'S NATIONAL WATER AGENCY

OUR MENTOR:

Mentor picture here:

Mentor Comments:…

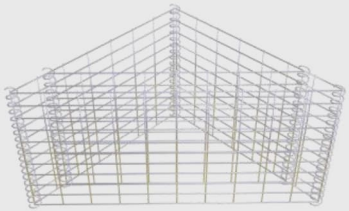

Q&A



APPENDICES

Flexible Rubber Tyre High-Strength Net Breakwater

Implementation Method

 **Concrete Posts**	 **Triangular Gabions**	 **Flexible High-Strength Nets**	 **Scrap Tyres**
Spaced every 5 to 6m, embedded into the seabed.	Placed between posts, dimensions 1m (H) x 1.5m (W) x 1m (L).	Stretched between gabions, mesh size 50mm to 100mm.	Diameter 0.58m, distributed within the net structure.

Solution	Initial	Price	Implementation	Price	Maintenance	Price	Total amount
Xbloc revetment with ecological enhancement (3.5 years)	Xbloc	60,000,000	Installation	20,000,000	Routine inspection	3,500,000	275,500,000
	Ecological items	30,000,000	Construction	75,000,000	Repairment	2,000,000	
	Rock sediments	15,000,000	Monitoring plant growth	30,000,000	Environment assessment	15,000,000	
	Machinery	2,000,000	Transportation	3,000,000			
	Consultation	4,000,000	Contingency	15,000,000			
	Site survey	1,000,000					
	Subtotal:	112,000,000	Subtotal:	143,000,000	Subtotal:	20,500,000	
Mangrove expansion with Tetrapot (2 years)							43,000,000
	Tetrapot	15,000,000	Installation	2,000,000	Routine inspection	2,500,000	
	Plants	1,000,000	Planting	5,000,000	Repairment	1,500,000	
	Machinery	1,000,000	Monitoring plant growth	5,000,000	Environment assessment	5,000,000	
	Engineering & design	1,000,000	Transportation	500,000			
	Consultation	1,000,000	Contingency	1,500,000			
	Site survey	1,000,000					
	Subtotal:	20,000,000	Subtotal:	14,000,000	Subtotal:	9,000,000	
Modular detention tanks (1 year)							37,700,000
	Modular detention tanks	22,000,000	Installation	1,500,000	Routine inspection	1,000,000	
	Geotextiles	300,000	Construction	3,500,000	Repairment	2,000,000	
	PVC pipes	800,000	Integration	3,500,000			
	Consultation	1,000,000	Testing	500,000			
	Site survey	400,000	Transportation	200,000			
			Contingency	1,000,000			
	Subtotal:	24,500,000	Subtotal:	10,200,000	Subtotal:	3,000,000	
Solar panel integrated floating breakwater (1 year)							37,400,000
	Solar panel	18,000,000	Installation	1,500,000	Routine inspection	2,000,000	
	Floating breakwater	3,000,000	Integration	1,000,000	Repairment	5,000,000	
	Marine machinery	800,000	Transportation	500,000	Utilities	1,000,000	
	Diving equipment	200,000	Testing	1,000,000			
	Consultation	1,500,000	Contingency	400,000			
	Engineering & design	1,500,000					
	Subtotal:	25,000,000	Subtotal:	4,400,000	Subtotal:	8,000,000	
Flexible High-Strength Net Tyre Breakwater (6 months)							1,580,000
	Recycled tyres	15,000	Installation	300,000	Routine inspection	200,000	
	Nets	15,000	Transportation	50,000	Repairment	100,000	
	Concrete	100,000	Testing	100,000	Environment assessment	200,000	
	Gabions	250,000	Contingency	50,000			
	Marine machinery	100,000					
	Engineering & design	50,000					
	Site survey	50,000					
	Subtotal:	580,000	Subtotal:	500,000	Subtotal:	500,000	
Ecological Jetty (4 months)							2,930,000
	Textured pilings	800,000	Installation	400,000	Routine inspection	300,000	
	Materials	500,000	Contingency	50,000	Repairment	200,000	
	Ecological items	300,000			Environment assessment	250,000	
	Consultation	50,000					
	Engineering & Design	80,000					
	Subtotal:	1,730,000	Subtotal:	450,000	Subtotal:	750,000	
Overall Project Subtotal:	Initial cost sum:	183,810,000	Implementation cost sum:	172,550,000	Initial cost sum:	41,750,000	$ 398,110,000.00

Sustainability of Solutions

Xbloc

Xbloc characteristic can be **applied at a steep slope reducing the need for rock material**.

Up to **50%** reduction in carbon





Tetrapot

Tetrapot produced by sustainable concrete (GGBS replacement) is more sustainable and also have **higher long term strength** than Ordinary Portland Cement concrete.

Up to **25%** strength increase !!

For example

1 Year Strength

GGBS 50% replacement: 36-45 MPa

OPC: 33-36 MPa

CO_2 Emission

1 ton of GGBS: 60 kg of CO_2

1 ton of OPC: 1 ton of CO_2

95% reduction in carbon emission

Sustainability Analysis

Eco Cost of Materials

Cost to mitigate environmental damage caused by the production and lifecycle of materials

- **GWP - Global Warming Potential**
- **AP - Acidification Potential**
- **EP - Eutrophication Potential**
- **ODP - Ozone Depletion Potential**
- **POCP - Photochemical Ozone Creation Potential**
- **ADP - Abiotic Depletion Potential**

Table 6. Comparison of environment impact between common and sustainable materials in Northwest Coast 1

	Common Material		Sustainable Material	
	Tetrapod	**Floating Breakwater**	**Tetrapod**	**Floating Breakwater**
Abiotic Depletion Potential	0.000015	0.12	0.075	0.045
Global Warming Potential	0.095	0.1925	0.0125	0.05
Ozone Depletion Potential	0.000015	0.0000047	0.000017	0.0000011
Acidification Potential	0.02	0.18	0.01	0.04
Photochemical Ozone Creation Potential	0.005	0.007	0.003	0.002
Eutrophication Potential	0.0025	0.0225	0.01	0.0045
Eco-cost of Materials (USD/Kg)	0.125	0.522	0.11	0.142

Table 8. Comparison of environment impact between common and sustainable materials in Northwest Coast 2

	Common Material	Sustainable Material
	Bio Revetment	**Bio Revetment**
Abiotic Depletion Potential	0.045	0.01
Global Warming Potential	0.05	0.01
Ozone Depletion Potential	0.0000011	0.00000022
Acidification Potential	0.04	0.02
Photochemical Ozone Creation Potential	0.002	0.001
Eutrophication Potential	0.0045	0.003
Eco-cost of Materials (USD/Kg)	0.142	0.044

Table 8. Comparison of environment impact between common and sustainable materials in Northwest Coast 2

	Common Material	Sustainable Material
	Bio Revetment	**Bio Revetment**
Abiotic Depletion Potential	0.045	0.01
Global Warming Potential	0.05	0.01
Ozone Depletion Potential	0.0000011	0.00000022
Acidification Potential	0.04	0.02
Photochemical Ozone Creation Potential	0.002	0.001
Eutrophication Potential	0.0045	0.003
Eco-cost of Materials (USD/Kg)	0.142	0.044

Table 4. List of seagrass species reported from Singapore, including historic and current records. Species include: CR, *Cymodocea rotundata*; CS, *Cymodocea serrulata*; HP*, *Halodule pinifolia*; HU, *Halodule uninervis*; SI, *Syringodium isoetifolium*; EA, *Enhalus acoroides*; HB, *Halophila beccarii*; HD, *Halophila decipiens*; HM*, *Halophila minor*; HO, *Halophila ovalis*; HS, *Halophila spinulosa*; TH, *Thalassia hemprichii*. NB: HP* and HM* are synonymous with HU and HO, respectively. Data sources: 1, (Yaakub et al., 2013); 2, herbaria only; 3, denotes species observed during field observations (2010–2014); 4, Loo et al., (1996); 5, Singapore herbarium; 6, Sinclair, (1953); 7, den Hartog, (1970); 8, Noiraksar et al., (2012); 9, www.flickr.com; 10, Chng & Chou, 2014.

Locations	Habitat	CR	CS	HP*	HU	SI	EA	HB	HD	HM*	HO	HS	TH
Western Johor Strait													
Tuas	estuary						1,9				1	4	
Lim Chu Kang Mangroves	estuary							1					
Sungei Buloh + Kranji	estuary						1	1					
Mandai	estuary							1					
Pulau Pergam	estuary						1		1	1	1	1	